

Partial Derivatives Problems with Solutions

Chain Rule and Applications

January 18, 2026

Problem 1

If $z = f\left(\frac{y}{x}\right) + \sqrt{x^2 + y^2}$, show that:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \sqrt{x^2 + y^2}$$

Solution: Let $u = \frac{y}{x}$. Then $z = f(u) + \sqrt{x^2 + y^2}$.

Compute partial derivatives:

$$\begin{aligned} \frac{\partial z}{\partial x} &= f'(u) \frac{\partial u}{\partial x} + \frac{x}{\sqrt{x^2 + y^2}} \\ &= f'(u) \left(-\frac{y}{x^2}\right) + \frac{x}{\sqrt{x^2 + y^2}} \end{aligned}$$

$$\begin{aligned} \frac{\partial z}{\partial y} &= f'(u) \frac{\partial u}{\partial y} + \frac{y}{\sqrt{x^2 + y^2}} \\ &= f'(u) \left(\frac{1}{x}\right) + \frac{y}{\sqrt{x^2 + y^2}} \end{aligned}$$

Now compute:

$$\begin{aligned} x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} &= x \left[f'(u) \left(-\frac{y}{x^2}\right) + \frac{x}{\sqrt{x^2 + y^2}} \right] \\ &\quad + y \left[f'(u) \left(\frac{1}{x}\right) + \frac{y}{\sqrt{x^2 + y^2}} \right] \\ &= f'(u) \left(-\frac{y}{x}\right) + \frac{x^2}{\sqrt{x^2 + y^2}} \\ &\quad + f'(u) \left(\frac{y}{x}\right) + \frac{y^2}{\sqrt{x^2 + y^2}} \\ &= \frac{x^2 + y^2}{\sqrt{x^2 + y^2}} = \sqrt{x^2 + y^2} \end{aligned}$$

The $f'(u)$ terms cancel, proving the result.

Problem 2

If $u = f(r, s, t)$ and $r = x/y$, $s = y/z$, $t = z/x$, prove that:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$$

Solution: Using the chain rule:

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial x} \\ &= \frac{\partial u}{\partial r} \left(\frac{1}{y} \right) + \frac{\partial u}{\partial s} (0) + \frac{\partial u}{\partial t} \left(-\frac{z}{x^2} \right) \\ &= \frac{1}{y} \frac{\partial u}{\partial r} - \frac{z}{x^2} \frac{\partial u}{\partial t}\end{aligned}$$

$$\begin{aligned}\frac{\partial u}{\partial y} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial y} \\ &= \frac{\partial u}{\partial r} \left(-\frac{x}{y^2} \right) + \frac{\partial u}{\partial s} \left(\frac{1}{z} \right) + \frac{\partial u}{\partial t} (0) \\ &= -\frac{x}{y^2} \frac{\partial u}{\partial r} + \frac{1}{z} \frac{\partial u}{\partial s}\end{aligned}$$

$$\begin{aligned}\frac{\partial u}{\partial z} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \frac{\partial t}{\partial z} \\ &= \frac{\partial u}{\partial r} (0) + \frac{\partial u}{\partial s} \left(-\frac{y}{z^2} \right) + \frac{\partial u}{\partial t} \left(\frac{1}{x} \right) \\ &= -\frac{y}{z^2} \frac{\partial u}{\partial s} + \frac{1}{x} \frac{\partial u}{\partial t}\end{aligned}$$

Now compute:

$$\begin{aligned}x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} &= x \left(\frac{1}{y} \frac{\partial u}{\partial r} - \frac{z}{x^2} \frac{\partial u}{\partial t} \right) \\ &\quad + y \left(-\frac{x}{y^2} \frac{\partial u}{\partial r} + \frac{1}{z} \frac{\partial u}{\partial s} \right) \\ &\quad + z \left(-\frac{y}{z^2} \frac{\partial u}{\partial s} + \frac{1}{x} \frac{\partial u}{\partial t} \right) \\ &= \frac{x}{y} \frac{\partial u}{\partial r} - \frac{z}{x} \frac{\partial u}{\partial t} - \frac{x}{y} \frac{\partial u}{\partial r} + \frac{y}{z} \frac{\partial u}{\partial s} \\ &\quad - \frac{y}{z} \frac{\partial u}{\partial s} + \frac{z}{x} \frac{\partial u}{\partial t} \\ &= 0\end{aligned}$$

All terms cancel, proving the result.

Problem 3

Given that $u(x, y, z) = f(x^2 + y^2 + z^2)$ where:

$$x = r \cos \theta \cos \phi, \quad y = r \cos \theta \sin \phi, \quad z = r \sin \theta$$

Find $\frac{\partial u}{\partial \theta}$ and $\frac{\partial u}{\partial \phi}$.

Solution: Let $w = x^2 + y^2 + z^2$. Then $u = f(w)$ and $\frac{\partial u}{\partial w} = f'(w)$.

First compute w in spherical coordinates:

$$\begin{aligned} x^2 + y^2 + z^2 &= (r \cos \theta \cos \phi)^2 + (r \cos \theta \sin \phi)^2 + (r \sin \theta)^2 \\ &= r^2 \cos^2 \theta \cos^2 \phi + r^2 \cos^2 \theta \sin^2 \phi + r^2 \sin^2 \theta \\ &= r^2 \cos^2 \theta (\cos^2 \phi + \sin^2 \phi) + r^2 \sin^2 \theta \\ &= r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2 \end{aligned}$$

So $w = r^2$ and $f'(w) = f'(r^2)$.

Now using chain rule:

$$\begin{aligned} \frac{\partial u}{\partial \theta} &= \frac{\partial u}{\partial w} \left(\frac{\partial w}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial \theta} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial \theta} \right) \\ &= f'(w) [2x(-r \sin \theta \cos \phi) + 2y(-r \sin \theta \sin \phi) + 2z(r \cos \theta)] \\ &= 2f'(r^2) [-r^2 \cos \theta \sin \theta \cos^2 \phi - r^2 \cos \theta \sin \theta \sin^2 \phi + r^2 \sin \theta \cos \theta] \\ &= 2f'(r^2) [-r^2 \cos \theta \sin \theta (\cos^2 \phi + \sin^2 \phi) + r^2 \sin \theta \cos \theta] \\ &= 2f'(r^2) [-r^2 \cos \theta \sin \theta + r^2 \sin \theta \cos \theta] = 0 \end{aligned}$$

Similarly:

$$\begin{aligned} \frac{\partial u}{\partial \phi} &= \frac{\partial u}{\partial w} \left(\frac{\partial w}{\partial x} \frac{\partial x}{\partial \phi} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial \phi} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial \phi} \right) \\ &= f'(w) [2x(-r \cos \theta \sin \phi) + 2y(r \cos \theta \cos \phi) + 2z(0)] \\ &= 2f'(r^2) [-r^2 \cos^2 \theta \cos \phi \sin \phi + r^2 \cos^2 \theta \sin \phi \cos \phi] = 0 \end{aligned}$$

Thus:

$$\boxed{\frac{\partial u}{\partial \theta} = 0 \quad \text{and} \quad \frac{\partial u}{\partial \phi} = 0}$$

This makes sense because u depends only on $r^2 = x^2 + y^2 + z^2$, which is independent of θ and ϕ in spherical coordinates.

Problem 4

Let $z = f(x, y)$, where $x = u^2 - v^2$ and $y = 2uv$. Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ in terms of partial derivatives of f .

Solution: Using the chain rule:

$$\begin{aligned} \frac{\partial z}{\partial u} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} \\ &= f_x \cdot 2u + f_y \cdot 2v \\ &= 2uf_x + 2vf_y \end{aligned}$$

$$\begin{aligned} \frac{\partial z}{\partial v} &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v} \\ &= f_x \cdot (-2v) + f_y \cdot 2u \\ &= -2vf_x + 2uf_y \end{aligned}$$

Thus:

$$\frac{\partial z}{\partial u} = 2uf_x + 2vf_y \quad \text{and} \quad \frac{\partial z}{\partial v} = -2vf_x + 2uf_y$$

Problem 5

If $u = f(x, y)$, $x = e^s \cos t$, $y = e^s \sin t$, show that:

$$\frac{\partial^2 u}{\partial s^2} + \frac{\partial^2 u}{\partial t^2} = e^{2s} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

Solution: First derivatives:

$$\begin{aligned} \frac{\partial x}{\partial s} &= e^s \cos t, & \frac{\partial x}{\partial t} &= -e^s \sin t \\ \frac{\partial y}{\partial s} &= e^s \sin t, & \frac{\partial y}{\partial t} &= e^s \cos t \end{aligned}$$

First partials of u :

$$\begin{aligned} \frac{\partial u}{\partial s} &= \frac{\partial u}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial s} \\ &= u_x e^s \cos t + u_y e^s \sin t \\ &= e^s (u_x \cos t + u_y \sin t) \end{aligned}$$

$$\begin{aligned} \frac{\partial u}{\partial t} &= \frac{\partial u}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial t} \\ &= u_x (-e^s \sin t) + u_y (e^s \cos t) \\ &= e^s (-u_x \sin t + u_y \cos t) \end{aligned}$$

Second derivative $\frac{\partial^2 u}{\partial s^2}$:

$$\begin{aligned} \frac{\partial^2 u}{\partial s^2} &= \frac{\partial}{\partial s} [e^s (u_x \cos t + u_y \sin t)] \\ &= e^s (u_x \cos t + u_y \sin t) \\ &\quad + e^s [(u_{xx} e^s \cos t + u_{xy} e^s \sin t) \cos t \\ &\quad + (u_{yx} e^s \cos t + u_{yy} e^s \sin t) \sin t] \\ &= e^s (u_x \cos t + u_y \sin t) \\ &\quad + e^{2s} [u_{xx} \cos^2 t + 2u_{xy} \cos t \sin t + u_{yy} \sin^2 t] \end{aligned}$$

Second derivative $\frac{\partial^2 u}{\partial t^2}$:

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} &= \frac{\partial}{\partial t} [e^s (-u_x \sin t + u_y \cos t)] \\ &= e^s [(-u_{xx} e^s \sin t + u_{xy} e^s \cos t)(-\sin t) \\ &\quad + (-u_x \cos t) + (u_{yx} e^s \sin t + u_{yy} e^s \cos t) \cos t - u_y \sin t] \\ &= e^s (-u_x \cos t - u_y \sin t) \\ &\quad + e^{2s} [u_{xx} \sin^2 t - 2u_{xy} \sin t \cos t + u_{yy} \cos^2 t] \end{aligned}$$

Now add them:

$$\begin{aligned}\frac{\partial^2 u}{\partial s^2} + \frac{\partial^2 u}{\partial t^2} &= e^s(u_x \cos t + u_y \sin t) - e^s(u_x \cos t + u_y \sin t) \\ &\quad + e^{2s} [u_{xx}(\cos^2 t + \sin^2 t) + u_{yy}(\sin^2 t + \cos^2 t)] \\ &= e^{2s}(u_{xx} + u_{yy})\end{aligned}$$

Thus:

$$\boxed{\frac{\partial^2 u}{\partial s^2} + \frac{\partial^2 u}{\partial t^2} = e^{2s} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)}$$

Problem 6: Computational Application

A rectangular metal plate has length $L = 50.0$ cm and width $W = 30.0$ cm. The coefficient of linear expansion is $\alpha = 1.2 \times 10^{-5}$ per $^\circ\text{C}$ for both dimensions.

(a) The area is $A = LW$. If temperature increases by $\Delta T = 10^\circ\text{C}$, use linear approximation to estimate ΔA .

(b) Calculate exact change and find percentage error.

(c) Explain why approximation is accurate.

Solution:(a) Linear approximation:

$$\Delta A \approx A_L \Delta L + A_W \Delta W = W \Delta L + L \Delta W$$

where $\Delta L = L\alpha\Delta T$ and $\Delta W = W\alpha\Delta T$.

Calculate:

$$\begin{aligned}\Delta L &= 50.0 \times 1.2 \times 10^{-5} \times 10 = 0.006 \text{ cm} \\ \Delta W &= 30.0 \times 1.2 \times 10^{-5} \times 10 = 0.0036 \text{ cm} \\ \Delta A &\approx 30.0 \times 0.006 + 50.0 \times 0.0036 \\ &= 0.18 + 0.18 = 0.36 \text{ cm}^2\end{aligned}$$

(b) Exact calculation:

$$\begin{aligned}L' &= L(1 + \alpha\Delta T) = 50.006 \text{ cm} \\ W' &= W(1 + \alpha\Delta T) = 30.0036 \text{ cm} \\ A' &= L'W' = 50.006 \times 30.0036 = 1500.5400216 \text{ cm}^2 \\ A &= 50.0 \times 30.0 = 1500.0 \text{ cm}^2 \\ \Delta A_{\text{exact}} &= 1500.5400216 - 1500.0 = 0.5400216 \text{ cm}^2\end{aligned}$$

Percentage error:

$$\text{Error} = \frac{0.5400216 - 0.36}{0.5400216} \times 100\% \approx 33.33\%$$

(c) The linear approximation neglects the second-order term $\Delta L \Delta W$:

$$\Delta A_{\text{exact}} = W \Delta L + L \Delta W + \Delta L \Delta W$$

Here $\Delta L \Delta W = 0.006 \times 0.0036 = 2.16 \times 10^{-5} \text{ cm}^2$, which is very small compared to the first-order terms. The approximation is accurate because α is very small (1.2×10^{-5}), making ΔL and ΔW very small relative to L and W .

Problem 7: Error Analysis

The period of a pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$, with $L = 1.00$ m, $\Delta L = \pm 0.01$ m, $g = 9.80$ m/s², $\Delta g = \pm 0.05$ m/s².

(a) Estimate maximum error in T . (b) Which variable contributes more? (c) Reduce error by 50

Solution:(a) Linear approximation:

$$\Delta T \approx \frac{\partial T}{\partial L} \Delta L + \frac{\partial T}{\partial g} \Delta g$$

First compute partial derivatives:

$$\frac{\partial T}{\partial L} = \frac{\pi}{\sqrt{gL}}, \quad \frac{\partial T}{\partial g} = -\frac{\pi\sqrt{L}}{g^{3/2}}$$

At given values:

$$\begin{aligned} T &= 2\pi\sqrt{\frac{1.00}{9.80}} \approx 2.006 \text{ s} \\ \frac{\partial T}{\partial L} &= \frac{\pi}{\sqrt{9.80 \times 1.00}} \approx 1.003 \text{ s/m} \\ \frac{\partial T}{\partial g} &= -\frac{\pi\sqrt{1.00}}{(9.80)^{3/2}} \approx -0.1025 \text{ s}^3/\text{m}^2 \end{aligned}$$

Maximum error (worst case):

$$\begin{aligned} \Delta T_{\max} &\approx \left| \frac{\partial T}{\partial L} \Delta L \right| + \left| \frac{\partial T}{\partial g} \Delta g \right| \\ &\approx 1.003 \times 0.01 + 0.1025 \times 0.05 \\ &\approx 0.01003 + 0.005125 = 0.015155 \text{ s} \end{aligned}$$

(b) Contribution from L : 0.01003 s Contribution from g : 0.005125 s

L contributes more to the error. This can also be seen from the relative sensitivities:

$$\frac{\Delta T}{T} \approx \frac{1}{2} \left(\frac{\Delta L}{L} - \frac{\Delta g}{g} \right)$$

Relative errors: $\frac{\Delta L}{L} = 1\%$, $\frac{\Delta g}{g} \approx 0.51\%$, so L has larger relative error.

(c) To reduce error by 50

$$\Delta L_{\text{new}} = \frac{\Delta L}{2} = 0.005 \text{ m}$$

Since L contributes about $\frac{2}{3}$ of the total error, reducing its error by half reduces total error by approximately $\frac{1}{3}$, close to 50

Alternatively, using exact proportions:

$$\text{New error} = 0.5 \times 0.015155 = 0.0075775$$

If we reduce only ΔL :

$$\begin{aligned} 0.1025 \times 0.05 + 1.003 \times \Delta L_{\text{new}} &= 0.0075775 \\ \Delta L_{\text{new}} &= \frac{0.0075775 - 0.005125}{1.003} \approx 0.002445 \text{ m} \end{aligned}$$

Reduction factor: $\frac{0.002445}{0.01} \approx 0.2445$, so reduce by factor of about 4.